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Subject: PREVENTION OF ELECTROCUTION CASUALTIES – SAFE USE OF PORTABLE ELECTRICAL EQUIPMENT, ELECTRICAL ISOLATION, AND COMPLIANCE WITH SAFETY MANAGEMENT SYSTEM REQUIREMENTS

Ref:

(a) SOLAS Chapters II-1 and II-2
(b) ISM Code – Sections 1.2.2, 6, 7, and 10
(c) IEC 60092 – Electrical Installations in Ships
(d) ILO Code of Practice: Accident Prevention on Board Ship at Sea and in Port
(e) Liberian Marine Notice INS-001 – Safety Inspections of Liberian Ships
(f) Liberian Marine Notice SAF-003 – Enclosed Space Safety & Procedures

Dear Shipowners/Operators/Masters/Engine Department Personnel:

The purpose of this Marine Advisory is to reinforce mandatory safety requirements and operational controls related to electrical hazards aboard Liberian-flagged vessels, with particular emphasis on the use, handling, relocation, and isolation of portable electrical equipment.

Recent very serious marine casualties involving fatal electrocution demonstrate that routine shipboard activities can present significant electrical risks when energized equipment is handled without adequate isolation, inspection, task-specific risk assessment, supervision, or appropriate safeguards.

This Advisory summarizes:

- Required Regulatory and Safety Management System (SMS) obligations
- Common deficiencies identified during casualty investigations
- Lessons learned from electrocution incidents
- Mandatory actions to prevent recurrence

This Advisory does not address investigation procedures, evidence collection, or reporting steps beyond basic regulatory obligations.

Electrocution Casualty Overview

The Administration has reviewed several fatal electrocution incidents that occurred:

- Both at sea and in port

- During routine operations such as cargo hold cleaning, cargo condition inspection, and night deck duties
- Involving portable electrical lighting equipment connected to shipboard power supplies

In all reviewed cases, the casualties involved direct or indirect contact with energized electrical equipment, frequently in conductive environments such as steel ladders, cargo holds, crane pedestals, or high-humidity conditions. Prompt emergency response actions were taken; however, all incidents resulted in fatalities.

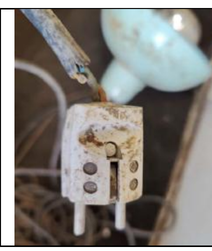
Marine Casualty Summaries:

Case 1 – Cargo Hold Cleaning

An Ordinary Seaman suffered fatal electrocution while carrying an energized portable light up a steel ladder during cargo hold cleaning operations. While ascending the ladder, the light struck the steel structure, shattering the protective glass enclosure and exposing live electrical components. The seafarer came into contact with the energized components while simultaneously grounded through the ship's structure, resulting in an electrical shock.

Case 2 – Cargo Condition Inspection

An Able Seaman collapsed while handling and adjusting a portable cargo light inside a cargo hold during a routine cargo condition inspection. The light was supplied via extension cables. A second crew member who attempted to assist also experienced an electric shock before the electrical supply was isolated. Emergency response actions were initiated after power was isolated.



Case 3 – Night Deck Duties

An Ordinary Seaman assigned to night deck duties was found unresponsive inside a crane pedestal, a space containing energized electrical systems. A portable light projector remained energized and connected to a 220-V socket at the time of discovery. The projector's electrical cable had a temporary taped repair, and the space contained exposed and energized electrical components, increasing the risk of electrical contact.

Key Safety Failures Identified

Investigations identified the following recurring deficiencies:

- Portable electrical equipment remained energized during handling or relocation
- Electrical tasks were treated as routine operations without task-specific risk assessment
- Use of portable lights in humid, wet, enclosed, or steel-structured environments

- Absence of a formal inspection, testing, and certification regime for portable electrical equipment
- Damaged or temporarily repaired electrical cables remaining in service
- Inadequate supervision of inexperienced personnel
- Entry into restricted or electrical spaces without authorization or permit
- Lack of appropriate electrical-grade PPE during handling of energized equipment

Flag Administration Requirements and Best Practices

The Administration has reviewed recent electrocution casualties and notes that non-compliance with established Safety Management System (SMS) requirements, including inadequate assessment and control of electrical hazards, remains a significant contributory factor.

In accordance with SOLAS, the ISM Code, and applicable Liberian Marine Notices, Shipowners, Operators, and Masters are reminded that hazard identification and risk assessment are mandatory requirements for all shipboard operations, including routine tasks involving portable electrical equipment.

Review and Implement Risk-Based Procedures

- Ensure the SMS adequately addresses electrical hazards associated with the use, handling, relocation, and isolation of portable electrical equipment.
- Require task-specific risk assessments for operations involving:
    - Portable electrical lighting and temporary equipment
    - Cargo holds, crane pedestals, and restricted or enclosed spaces
    - Humid, wet, or high-conductivity environments
    - Night work or operations performed with limited supervision
- Risk assessments shall clearly identify:
    - Electrical isolation and disconnection requirements
    - Potential grounding paths through ship structures
    - Environmental factors increasing electrocution risk
    - PPE, supervision, and authorization requirements

The Administration emphasizes that classifying electrical work as a “routine task” does not remove the obligation to formally assess and control associated risks.

Portable Electrical Equipment Control

- Establish and maintain a portable electrical equipment register within the planned maintenance system.
- Implement regular inspection and testing of portable lights, cables, and connectors.
- Remove from service any equipment with damaged housings, lenses, defective connectors, or temporary or taped cable repairs.

Electrical Isolation and Safe Work Practices

- Ensure all portable electrical equipment is de-energized and unplugged prior to:
  - Relocation or transport
  - Climbing ladders or accessing elevated positions
  - Adjustment, dismantling, or removal
- Verify electrical isolation prior to recovery or emergency response actions.

Supervision, Training, and Oversight

- Prohibit inexperienced personnel from handling energized electrical equipment without direct supervision.
- Reinforce crew training on:
  - Electrical shock hazards and conductive shipboard structures
  - Increased risks associated with moisture and humidity
  - Safe ladder use and three-point contact principles
- Include electrical hazard management and portable-equipment control in onboard audits, drills, inspections, and toolbox meetings.

Required Actions to Prevent Recurrence

1. Prohibit carrying/relocation of energized portable electrical equipment.
2. Implement routine inspection and insulation testing for all portable electrical equipment.
3. Eliminate temporary or taped electrical repairs.
4. Provide appropriate electrical PPE where exposure risks exist.
5. Treat crane pedestals, electrical rooms, and similar spaces with the power supply sources as restricted electrical spaces requiring authorization.
6. Maintain records of inspections, deficiencies, corrective actions, and equipment removed from service.

The Administration reminds all operators that prevention of electrocution is a fundamental safety obligation. Strict compliance with electrical isolation, risk-assessment, inspection, supervision, and training requirements is essential to prevent recurrence of fatal electrical incidents.

For more information, please contact the Investigations Department at investigations@liscr.com.

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